

MLOps Foundations*SRAI Book 5 MLOps and AI Engineering | PU-B05-C01 | Controlled Lesson v0.1**Mbaye Kebe*

This controlled lesson explains how an experimental model becomes a reproducible, deployable, observable, governable and reversible service. Its executable thread is a synthetic agricultural-risk classifier, while its institutional applications span Agriculture, Education, Health and Habitat. The chapter emphasizes transferable evidence and accountable decisions rather than a particular vendor platform.

Control	Value
Canonical notebook	V5_N01_mlops_foundations.ipynb
Reconstruction status	Developed from first principles; original Chapter 1 contained no usable manuscript
Notebook seed	42
Data status	Synthetic educational data
Release posture	Fail closed until named governance approval

Reader guidance

Read the lesson sequentially once, then use the lifecycle, evidence and responsibility tables as operational references. Run the companion notebook after Sections 5 through 12. Every numerical result must be tied to the exact data, configuration, environment and code state that produced it.

Learning objectives

- Explain why model development and model operations are one governed lifecycle.
- Specify a reproducibility contract and an evidence chain.
- Design acceptance gates for data, models, services and release.
- Interpret monitoring and drift without treating alerts as proof of causality.
- Distinguish technical readiness from authorized production use.
- Design rollback, incident response and retirement controls.

Chapter map

Part	Purpose	Notebook evidence
Foundations	Lifecycle, risk and operating model	Configuration and environment
Build	Data, splits, baseline and candidate	Accepted data and metrics
Control	Registry, contracts and gates	Manifest and hashes
Operate	Monitoring, incident and rollback	PSI, evidence inventory and rollback record

1 The production problem

A model can be statistically strong and operationally unsafe. Development evidence answers whether a candidate learned a useful relationship under a declared information boundary. Production evidence must also establish that the correct artifact receives valid inputs, returns a stable output contract, remains observable, can be reversed and is used only for an authorized purpose.

In the agricultural-risk case, a notebook may report acceptable recall, yet a service can still fail because rainfall units changed, a category was renamed, credentials expired or an intervention program used the score for an unauthorized eligibility decision. MLOps makes these dependencies explicit and testable.

The central object is therefore not the model file. It is a governed system containing data contracts, transformations, configuration, code, environment, tests, approvals, deployment state, monitoring rules, incident procedures and retirement criteria.

Failure mode	Why notebook success is insufficient	Required control
Schema drift	Training data contract may no longer hold	Production data validation
Training serving skew	Transformations differ	Bound preprocessing pipeline
Silent degradation	No outcome feedback	Layered monitoring
Unauthorized use	Metrics do not define legal purpose	Named use approval
Irreversible change	Old model alone may not restore service	Complete rollback manifest

2 MLOps as an evidence discipline

MLOps combines machine-learning practice, software engineering, data engineering, platform operations and governance. Automation is valuable because it makes repeated controls cheaper and more consistent. Automation is dangerous when teams confuse a successful pipeline with authority to deploy.

A defensible lifecycle produces evidence at every transition. Data acceptance precedes training. Evaluation precedes registry entry. Independent review precedes promotion. Deployment verification precedes traffic expansion. Monitoring precedes retraining or rollback decisions. Each transition has a named owner and a declared failure response.

The SRAI position is deliberately strict: technical completion is evidence, not authorization. A pipeline may recommend that a candidate is eligible for review; it must not manufacture the organizational authority that approves consequential use.

AuthorizedRelease

*= TechnicalEvidence $\wedge$ Security $\wedge$ Monitoring $\wedge$ Rollback
 $\wedge$ GovernanceApproval*

3 Lifecycle states and transitions

A useful lifecycle distinguishes problem framing, data preparation, experimentation, validation, registration, approval, deployment, monitoring, change and retirement. These are controlled states rather than a loose sequence of tasks. Every transition identifies inputs, outputs, decision criteria and accountable roles.

Iteration does not disappear. A monitoring alert can return work to data investigation, model development or policy review. An incident can trigger rollback before the cause is fully understood. Retirement may be required when the purpose expires even if the system remains technically functional.

State must be durable. If the orchestration service restarts, the team must still know which data were accepted, which model was approved, which version is serving and which rollback target is valid.

State	Entry evidence	Exit decision
Experiment	Accepted development data	Candidate selected
Validated	Independent metric and slice review	Registry eligible
Registered	Immutable artifact and lineage	Promotion review
Deployed	Approved manifest and service checks	Bounded traffic
Monitored	Telemetry and outcome evidence	Continue, investigate, retrain or rollback
Retired	Purpose ended or risk unacceptable	Archive and revoke

4 Roles and segregation of duties

Model builders optimize candidates; data owners attest provenance and permitted use; platform operators control runtime; independent evaluators challenge evidence; domain owners judge consequences; governance authorities approve use. Small teams may combine jobs, but they must preserve distinct decisions and an auditable record of who built, reviewed and authorized.

Applied Agriculture lens. Before this control is accepted, the team must name the decision supported, the population and time period covered, the data owner, the operational user and the person authorized to stop the system. Evidence should show both the expected path and at least one deliberately injected failure. A successful demonstration on synthetic or historical data remains bounded to that evidence; it does not establish prospective validity, lawful authority or safe use in a different institution.

Review questions. What exact artifact proves the control ran? Which mandatory failure blocks the next state? Who investigates the failure, who may approve an exception, and when does that exception expire? What evidence would allow an independent reviewer to reconstruct the decision six months later?

- Evidence must be versioned and attributable.
- Mandatory failures block the transition.
- Exceptions require an owner, rationale and expiry.
- The control must specify what happens next.

5 Reproducibility contract

A reproducibility contract names code revision, data snapshot, schema, configuration, dependency lock, random seed, hardware assumptions and tolerance. Reproducibility is a controlled reconstruction claim, not a vague statement that the notebook ran once. Seed 42 is preserved for the canonical notebook, while the lesson explains that deterministic seeds do not eliminate platform and library variation.

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$$EvidenceID = H(Data) || H(Code) || H(Config) || H(Environment)$$

Here H denotes a cryptographic digest and the concatenation symbolizes a manifest, not a claim that hashes capture meaning. The manifest must also record schema, provenance, intended use and ownership.

6 Data contracts and acceptance

A data contract defines fields, types, units, ranges, category policy, time semantics, missingness and ownership. Acceptance tests should fail before training when a mandatory condition is violated. Exceptions require an owner, rationale, expiration and impact analysis; silently coercing invalid data destroys the evidence chain.

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7 Information boundaries and splits

Training, validation and test partitions represent an information boundary. Temporal or entity leakage can yield impressive but indefensible results. Split membership, grouping logic and timestamps belong in the manifest. The test set is not a repeatedly consulted optimization resource.

Applied Habitat lens. Before this control is accepted, the team must name the decision supported, the population and time period covered, the data owner, the operational user and the person authorized to stop the system. Evidence should show both the expected path and at least one deliberately injected failure. A successful demonstration on synthetic or historical data remains bounded to that evidence; it does not establish prospective validity, lawful authority or safe use in a different institution.

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8 Baselines and experiment tracking

A baseline creates a minimum evidence threshold. Experiment tracking binds parameters, metrics, artifacts and provenance to a run identifier. Comparability requires the same evaluation population, metric definitions and threshold policy. A dashboard of incomparable runs is decoration, not evidence.

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9 Evaluation and uncertainty

Accuracy alone is rarely sufficient. Precision, recall, F1, ROC AUC, calibration and slice metrics address different questions. Threshold selection must reflect consequences and resource constraints. Estimates from finite samples carry uncertainty, and operational performance can differ after deployment.

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$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Metric	Operational interpretation	Risk if isolated
Recall	Share of true high-risk cases flagged	May overwhelm intervention capacity
Precision	Share of flags that are true high risk	May miss many high-risk cases
ROC AUC	Ranking discrimination across thresholds	Does not fix a decision threshold
Calibration	Agreement between probability and frequency	Can vary by subgroup and time

10 Versioning and lineage

Version code, data, configuration, environments and model artifacts. Mutable labels such as latest are convenient pointers, not identities. Content digests protect byte identity; semantic versions communicate intended compatibility; lineage connects every release decision to the evidence that supported it.

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11 Testing the ML system

A layered portfolio includes unit, integration, schema, statistical, model, contract, security, performance and recovery tests. Model tests use tolerances because some computations vary. A passing average must not conceal a failed mandatory slice, safety or rollback test.

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Layer	Example control	Failure response
Data	Schema and range tests	Block ingestion
Software	Unit and integration tests	Block build
Model	Metric and slice thresholds	Reject candidate
Service	Contract and load tests	Block deployment
Operations	Rollback drill	Block promotion
Governance	Named approval	Remain unapproved

12 Registry and promotion

A model registry records versions, lineage, evaluation, lifecycle status and ownership. Registration is not deployment. Promotion should reference an immutable manifest and preserve an approval record. Production systems should reject an artifact whose digest or dependencies differ from the approved candidate.

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13 Serving and API contracts

A serving contract declares required inputs, units, allowed values, response fields, error behavior, version identity and latency objectives. Health and readiness are distinct: a process can be alive while unable to serve valid predictions. Backward compatibility and deprecation must be explicit.

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14 Deployment strategies

Shadow deployment observes behavior without affecting decisions. Canary deployment exposes bounded traffic. Blue green deployment preserves two environments for controlled switching. Batch scoring suits scheduled decisions. Strategy selection depends on reversibility, latency, consequence and infrastructure—not fashion.

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15 Monitoring and drift

Monitor service health, data quality, feature distributions, predictions, outcomes, subgroup performance, security and cost. Drift is a change signal. It does not prove degradation or identify a cause. Alerts need thresholds, owners, severity, runbooks and suppression rules.

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$$PSI = \sum_i (a_i - e_i) \ln(a_i / e_i)$$

The notebook uses quantile bins, clips empty proportions for numerical stability and treats PSI as a diagnostic signal. Thresholds must be calibrated to the feature, volume and consequence of the application.

16 Incidents and rollback

Rollback restores a complete known-good operating state: model, preprocessing, schema, configuration, dependencies and routing. Teams should rehearse rollback before release. Incident records preserve detection, containment, impact, decision, recovery and lessons without waiting for perfect causal certainty.

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17 Retraining and change control

Retraining is a controlled change, not an automatic cure for drift. New data may encode changed policy, measurement error or inequity. A retrained candidate returns through acceptance, evaluation and approval gates. Automation can trigger investigation or candidate construction, but not consequential self-approval.

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18 Security privacy and cost

Protect credentials, artifacts, endpoints and logs. Minimize personal and sensitive data, enforce purpose limitation and retention, and prevent training data from leaking through diagnostics. Track compute, storage, network and human-review costs because an unaffordable control is unlikely to persist.

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Decision	Technical evidence	Authority question
Register	Candidate meets declared evaluation gates	Who may admit it to controlled review?
Deploy	Service and rollback tests pass	Who may authorize this use?
Retrain	Monitoring indicates material change	Who approves new data and purpose?
Retire	Purpose or supportability ended	Who confirms downstream closure?

19 Governance and institutional use

Governance connects technical evidence to mandate, law, ethics and accountability. Public institutions require clear authority, accessible explanations, contestability where decisions affect people, procurement controls and records that survive staff turnover. This applies to agricultural early warning, education support, health triage and habitat planning. The authorized use must be narrower than whatever the model can technically compute.

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20 Retirement

Retire a system when its purpose ends, evidence is no longer adequate, dependencies cannot be supported or risk exceeds benefit. Retirement includes traffic removal, credential revocation, archive, retention decisions, stakeholder communication and confirmation that downstream consumers no longer depend on the service.

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21 Worked case agricultural risk service

A ministry analytics team develops a synthetic classifier to demonstrate how drought-related production risk might be screened. The educational model cannot be used for credit, insurance, benefit allocation or farm-level action. Its purpose is to teach the evidence chain.

Stage	Evidence	Decision owner
Frame	Educational purpose and prohibited uses	Domain owner
Accept data	Schema, ranges, missingness and digest	Data owner
Select candidate	Baseline, validation and slice metrics	Independent evaluator
Register	Model digest, manifest and model card	Registry custodian
Promote	Security, monitoring, rollback and approval	Authorized governance body
Operate	Telemetry, outcomes and incident records	Service owner

The notebook deliberately sets governance approval to false. Even when model metrics pass, the final release decision remains blocked. This makes the distinction between evidence production and authority executable rather than rhetorical.

22 Cross sector transfer

The lifecycle remains stable across sectors, but evidence thresholds, harms, authority and response time change. A technically identical scoring service can therefore require a different operating model when it moves from planning analysis to an individual-level decision.

Sector	Illustrative use	Critical evidence	Required safeguard
Agriculture	Seasonal production-risk screening	Weather provenance, geographic coverage, drift	Do not convert an educational score into credit or benefit eligibility
Education	Early support identification	Attendance definitions, cohort leakage, subgroup recall	Human review and no punitive use of support indicators
Health	Clinical or public-health triage support	Clinical validation, calibration, safety and latency	Qualified clinical authority, escalation and conservative failure
Habitat	Housing condition or settlement-risk prioritization	Spatial resolution, survey quality, temporal validity	Community accountability, contestability and protection from exclusion

Cross-sector reuse is permitted only after reframing the purpose, data contract, evaluation population, risk analysis, governance authority and monitoring plan. Reusing code does not transfer validity or authorization.

23 Worked numerical example

Suppose validation contains 200 high-risk and 300 lower-risk cases. A threshold yields 160 true positives, 40 false negatives and 80 false positives. Precision is 160 divided by 240, or 0.667. Recall is 160 divided by 200, or 0.800. The F1 score is approximately 0.727.

$$Precision = 160 / (160 + 80) = 0.667$$

$$Recall = 160 / (160 + 40) = 0.800$$

$$F1 = 2(0.667)(0.800) / (0.667 + 0.800) = 0.727$$

Whether this threshold is acceptable depends on the authorized use, intervention capacity and consequences. The calculation supplies evidence; it does not decide policy.

24 Notebook laboratory

Run V5_N01_mlops_foundations.ipynb from top to bottom. Confirm that the environment record, configuration, synthetic data, acceptance results, split summary, metrics, model digest, contract tests, drift plot, release gates, model card, rollback record and evidence inventory are created.

- Do not change seed 42 for canonical reproduction.
- Report metrics to three decimals and record any platform differences.
- Inspect every failed gate before changing a threshold.
- Keep governance approval false during the first canonical execution.
- Treat the generated ZIP as evidence from an educational run, not a production release.

Artifact	Question
data_acceptance.csv	Which mandatory assumptions were tested?
run_record.json	Can the candidate result be reconstructed?
regional_slice_metrics.csv	Does aggregate performance hide a weak region?
release_manifest.json	Which gate blocks authorization?
rollback_record.json	Does rollback restore the complete contract?

25 AI Reasoning Clinic

A team reports that its pipeline passed, ROC AUC exceeded the threshold and the container is healthy. It asks an AI assistant to approve deployment. The correct response is to refuse the implied authorization and identify missing evidence: approved purpose, independent review, security controls, monitoring ownership, rollback rehearsal and named governance approval.

Reasoning pattern: separate observed facts from claims; map each claim to evidence; identify the decision authority; test mandatory gates conjunctively; state uncertainty; specify remediation; preserve the blocked state until authority acts.

SRAI Principle No. 141

An MLOps pipeline may assemble and verify technical evidence, but it cannot confer the institutional authority required for consequential production use.

26 Practical exercises

Exercise 1 Lifecycle map. Draw the lifecycle for a monthly agricultural-risk service. Name every artifact, gate, owner and return path.

Exercise 2 Data contract. Specify fields, units, ranges, time semantics and failure actions for rainfall, soil index and region.

Exercise 3 Evaluation. Compare two thresholds using precision, recall and intervention capacity. Defend one without claiming universal optimality.

Exercise 4 Lineage. Construct a manifest connecting dataset, split, code, environment, model and decision records.

Exercise 5 Monitoring. Design service, data, prediction, outcome, subgroup, security and cost indicators with owners and severity.

Exercise 6 Incident. Write a runbook for a sudden unit change from millimetres to centimetres.

Exercise 7 Governance. Explain why technical readiness cannot substitute for authorized use.

Exercise 8 Retirement. Design evidence that proves downstream consumers stopped using a retired endpoint.

27 Student assessment

Total: 100 marks. Recommended time: 4–6 hours plus notebook execution. Submit written reasoning, the executed notebook and generated evidence. Preserve seed 42, show intermediate calculations, report metrics to three decimals and use absolute tolerance 10^{-8} unless otherwise specified.

Question	Marks	Required response
1 Lifecycle and operating model	12	Define the governed lifecycle and assign accountable roles.
2 Reproducibility contract	12	Specify the evidence required to reconstruct a declared result.
3 Data acceptance	12	Design six mandatory checks and failure actions.
4 Evaluation	16	Calculate and interpret precision, recall and F1 from a confusion matrix.
5 Versioning and registry	12	Explain identity, lineage, registration and promotion.
6 Serving and monitoring	14	Design contract tests, indicators and alert ownership.
7 Incident and rollback	10	Provide a bounded response and complete rollback manifest.
8 Governance and notebook evidence	12	Explain the blocked release and propose authorized remediation.

28 Instructor solutions and marking guidance

1 Lifecycle and operating model

A complete answer addresses: Define the governed lifecycle and assign accountable roles. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 12 marks.

2 Reproducibility contract

A complete answer addresses: Specify the evidence required to reconstruct a declared result. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 12 marks.

3 Data acceptance

A complete answer addresses: Design six mandatory checks and failure actions. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 12 marks.

4 Evaluation

A complete answer addresses: Calculate and interpret precision, recall and F1 from a confusion matrix. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 16 marks.

5 Versioning and registry

A complete answer addresses: Explain identity, lineage, registration and promotion. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 12 marks.

6 Serving and monitoring

A complete answer addresses: Design contract tests, indicators and alert ownership. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 14 marks.

7 Incident and rollback

A complete answer addresses: Provide a bounded response and complete rollback manifest. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 10 marks.

8 Governance and notebook evidence

A complete answer addresses: Explain the blocked release and propose authorized remediation. It distinguishes technical evidence from decision authority, names artifacts and owners, explains failure

behavior and preserves traceability. Award approximately 40 percent for correct concepts, 30 percent for applied reasoning, 20 percent for evidence specificity and 10 percent for clarity. Maximum 12 marks.

29 Implementation checklist

Control	Minimum evidence	Status
Purpose	Approved intended and prohibited uses	☐
Data	Contract, provenance, acceptance and digest	☐
Experiment	Baseline, split, run and slice evidence	☐
Artifact	Immutable model and environment identities	☐
Service	Contract, security, load and readiness tests	☐
Monitoring	Indicators, thresholds, owners and runbooks	☐
Recovery	Tested rollback and incident process	☐
Governance	Named approval and expiry/review conditions	☐
Retirement	Revocation, archive and downstream confirmation	☐

30 Glossary

Term	Definition
Artifact	A versioned file or record produced or consumed by the lifecycle.
Data contract	Declared schema, semantics, quality rules and ownership for data exchange.
Drift	Change in a monitored distribution or relationship relative to a named reference.
Lineage	Trace connecting sources, transformations, runs, artifacts, deployments and decisions.
Model registry	Controlled record of model identities, evidence, lifecycle states and ownership.
Promotion gate	A condition and authorized decision required to move an artifact to a new state.
Readiness	Evidence that a service can accept valid traffic; distinct from process liveness.
Reproducibility	Ability to reconstruct a declared result from identified inputs and tolerances.
Rollback	Restoration of a complete known-good operating manifest.
Training-serving skew	Difference between training and production feature semantics or transformations.

31 Selected references

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3. Mitchell, M. et al. Model Cards for Model Reporting. FAT*, 2019.
4. NIST. Artificial Intelligence Risk Management Framework AI RMF 1.0. NIST AI 100-1, 2023.
5. Sculley, D. et al. Hidden Technical Debt in Machine Learning Systems. NeurIPS, 2015.
6. ISO IEC 42001 Artificial intelligence management system standard.
7. MLflow Documentation. Tracking and Model Registry.
8. DVC Documentation. Data and pipeline versioning.
9. Docker Documentation. Image build and security practices.
10. FastAPI Documentation. Request models, dependencies and deployment.

References support the control concepts in this lesson. Product documentation changes over time; implementation teams must pin the versions they actually use and archive relevant configuration evidence.

32 Education application

An education ministry develops an early-support indicator using attendance, assessment and school-context data. The authorized purpose is to help trained staff prioritize supportive outreach, not to punish learners, rank teachers or deny progression. The data contract must distinguish absence, transfer, delayed reporting and genuine dropout risk. Cohort and school leakage require explicit tests.

Evaluation must include recall, precision, calibration and subgroup slices by geography, grade, sex where lawful, disability status where ethically and legally justified, and school type. A flag should lead to human review and supportive verification. Monitoring covers data delays, category shifts, intervention capacity and whether flagged learners actually receive assistance. Appeals and correction paths are part of the operating model.

Gate	Education evidence	Block condition
Data	Attendance definitions and reporting delay	Unresolved school-system mismatch
Model	Cohort-safe validation and subgroup results	Material unsupported subgroup
Use	Approved support workflow	Punitive or unauthorized purpose
Operation	Named case-review and correction process	No accountable human review

33 Health application

A health program considers a triage-support model. Clinical consequence changes the evidence threshold: development metrics cannot replace clinical validation, qualified oversight, escalation rules and conservative failure behavior. Data provenance must distinguish routine care, surveillance and research. Missingness can be informative and must not be silently imputed without clinical reasoning.

The service contract specifies units, valid ranges, unavailable-result behavior and a route for urgent cases. Monitoring includes calibration, false-negative review, latency, downtime and distribution change across facilities. Deployment begins in shadow or tightly bounded use. A model output is advisory unless the authorized clinical workflow explicitly defines otherwise, and staff must know how to disregard or escalate it.

Risk	Required control	Evidence
False reassurance	Minimum sensitivity and escalation	Clinical validation report
Unit mismatch	Typed fields and range tests	Rejected malformed request
Unequal performance	Facility and population slices	Prospective monitoring
Unsafe outage	Conservative fallback	Downtime drill

34 Habitat application

A habitat agency uses spatial, survey and administrative data to prioritize inspection of housing conditions or settlement infrastructure. Spatial resolution and update frequency determine what the model can support. A neighborhood-level signal must not be presented as proof about an individual household. Survey nonresponse and inaccessible areas can create systematic blind spots.

The lifecycle records geospatial lineage, coordinate reference systems, imagery dates, survey design, aggregation rules and downstream map versions. Governance must protect communities from stigmatization, exclusion or enforcement uses outside the approved purpose. Contestability requires a way to correct records and document local evidence that the model could not observe.

Evidence object	Habitat question
Spatial manifest	Which imagery, boundaries and coordinate systems were used?
Coverage report	Which settlements or households were underrepresented?
Map contract	At what geographic level is interpretation valid?
Use approval	Can the output prioritize support without becoming an exclusion tool?

35 Agriculture application

An agricultural early-warning workflow combines weather, crop, market and survey information. Temporal alignment is critical: a feature unavailable at the prediction date creates leakage. Geographic averages can hide production-zone variation. Changes in instruments, satellite products or administrative boundaries require versioned transformations and back-testing.

The output should be connected to an explicit response protocol, such as additional field verification or resource planning. It must not silently become a farm-level credit, insurance or benefit decision.

Monitoring covers data arrival, spatial coverage, rainfall and price distributions, regional performance, intervention capacity and observed outcomes when they become available.

Lifecycle state	Agricultural evidence
Data accepted	Time-stamped sources, units, coverage and survey metadata
Candidate validated	Temporal holdout, regional slices and baseline comparison
Use approved	Named planning purpose and prohibited individual decisions
Monitored	Seasonal drift, data latency and outcome review

36 Cross sector evidence matrix

The same technical component can carry different consequence across sectors. The matrix below prevents a common portability error: assuming that reusable code transfers data validity, decision authority or acceptable risk.

Control	Agriculture	Education	Health	Habitat
Primary boundary	Season and geography	Cohort and school	Patient and facility	Household and spatial unit
High-cost error	Missed production stress	Missed learner support	Missed urgent condition	Missed unsafe housing
Human authority	Program or domain owner	Qualified education team	Qualified clinical authority	Housing or planning authority
Critical monitoring	Seasonal and spatial drift	Cohort and reporting shifts	Calibration and safety events	Coverage and boundary changes
Contestability	Field verification	Record correction and review	Clinical escalation	Community and household correction

A controlled transfer repeats problem framing, contract design, evaluation, governance and monitoring. Only generic engineering components—such as hashing, logging or test harnesses—may transfer directly, and even those require environment and security validation.

37 Reference MLOps architecture

The reference architecture separates evidence creation from authority. Source systems feed an accepted data zone. Versioned pipelines create features and candidate models. Evaluation services produce metric, slice and test evidence. A registry stores immutable candidates and lineage. An approval service records authorized decisions. Deployment tooling serves only the approved manifest. Monitoring and incident systems return evidence to owners.

Layer	Components	Control objective
Source and intake	Operational systems, surveys, sensors, files	Provenance and accepted schema
Build	Pipelines, feature code, training environment	Reproducible candidate
Evaluate	Baselines, metrics, slices, tests	Independent evidence
Control	Registry, manifest, approval record	Immutable authorized identity
Serve	Batch or online runtime, contracts	Reliable bounded inference
Observe	Logs, metrics, outcomes, incidents	Detect change and enable response

Security and governance cross every layer. Secrets, access, retention, purpose and ownership cannot be added only at the deployment edge. The architecture may be implemented with different products; the evidence contract remains the stable design object.

38 Release decision record

A release decision should be short enough to review and complete enough to audit. It identifies the candidate and every dependency by immutable reference; states intended and prohibited uses; summarizes evaluation and known limitations; records security, privacy, monitoring and rollback readiness; names the approver; and defines expiry or review conditions.

Decision field	Required content
Identity	Model, data, code, configuration and environment digests
Evidence	Evaluation population, metrics, slices, tests and limitations
Use	Authorized purpose, users, population and prohibited uses
Operations	Service objective, monitoring owner, alert route and rollback target
Authority	Reviewer, approver, decision date and conditions
Review	Expiry, change triggers and retirement criteria

The companion notebook generates a release manifest but leaves governance approval false. To authorize a real release, an organization would need a separate authenticated decision record. Editing a Boolean in a notebook is not governance.

39 Transition to Lesson 2

Lesson 1 established the complete evidence lifecycle. Lesson 2 focuses on reproducibility and environment management: dependency locking, runtime identity, deterministic reconstruction, tolerance, data snapshots and evidence that a result can be rebuilt outside the original workstation.

Production record

Field	Record
Production unit	PU-B05-C01
Lesson	MLOps Foundations
Canonical notebook	V5_N01_mlops_foundations.ipynb
Lesson version	v0.1 reconstruction candidate
Notebook posture	Independently runnable; synthetic data; seed 42
Authorization posture	Canonical run remains blocked pending governance approval
Owner	Mbaye Kebe

After pagination-affecting edits in Microsoft Word, refresh fields with Ctrl+A followed by F9. Validate the lesson and notebook together because the prose explains the control logic and the notebook produces executable evidence.